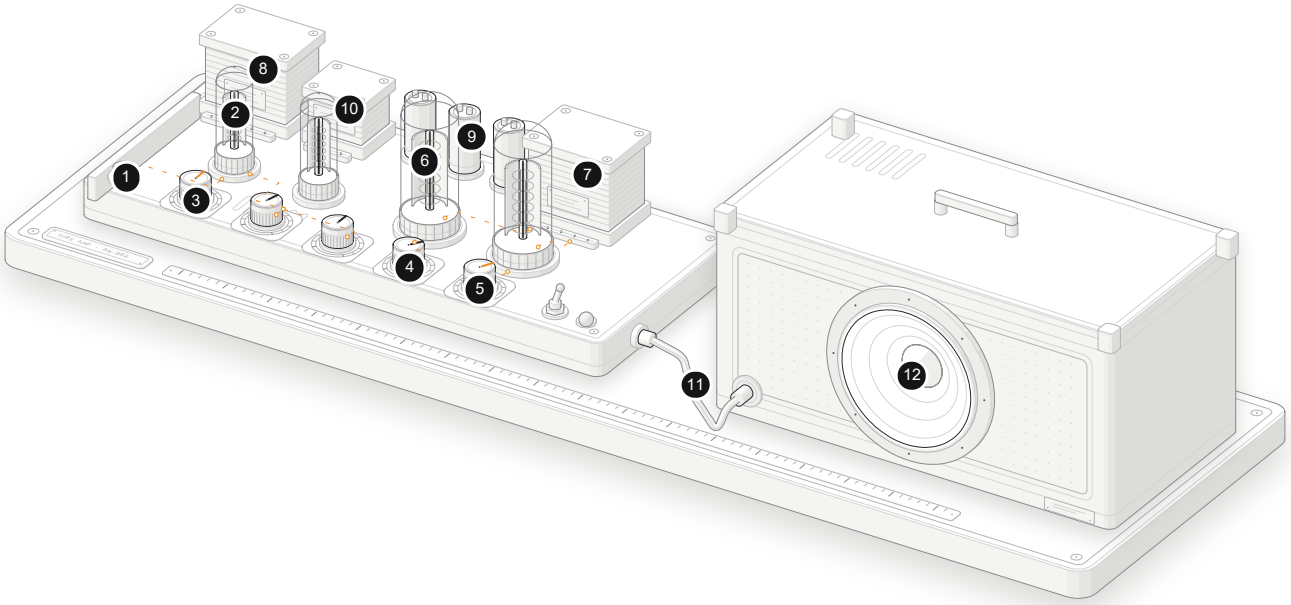
Tube guitar amp

AN-004
HOW IT WORKS
THREE TUBES, ONE CURVE



Why does a tube amp distort when you turn it up?

A guitar pickup makes about 0.3 V. Each tube stage multiplies the signal, but a tube can only swing so far, so it rounds off the tops of loud waves instead of cutting them. Turn Gain up and the second preamp tube runs out of headroom first, then the power tubes. The speaker cone follows whatever shape comes out.

0.3 V
FROM THE PICKUP

3 tube stages
GAIN AND SOFT-CLIP

7 %
DISTORTION AT GAIN 5

29 %
DISTORTION AT GAIN 10

01 **Input jack**
Where the guitar plugs in.

02 **Preamp tubes V1, V2**
Small tubes that raise the 0.3 V signal and start to clip.

03 **Gain knob**
A pot between V1 and V2 that sets how hard V2 is driven.

04 **Tone knobs**
Bass, mid and treble pots in the passive tone stack.

05 **Master knob**
Sets how hard the power tubes are driven.

06 **Power tubes V3, V4**
Large tubes that give the signal enough current for a speaker.

07 **Output transformer**
Steps the high-voltage tube signal down to drive the speaker.

08 **Power transformer**
Makes the high voltage the tubes run on.

09 **Filter capacitors**
Smooth the supply so the amp does not hum.

10 **Choke**
Helps filter the supply.

11 **Speaker cable**
Carries the output to the cabinet.

12 **Speaker**
The cone moves with the clipped wave.